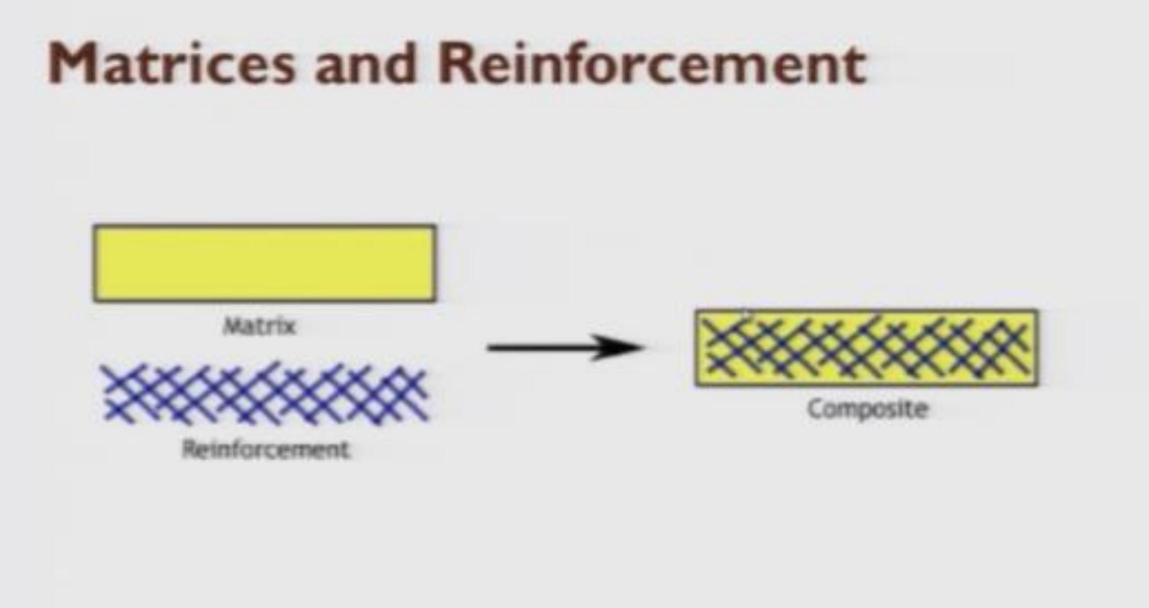
Composites: definition

- A broad definition of composite is: composed of more than one component, chemically distinct materials which when combined have improved properties over the individual materials, wherein those components are in separate phase.
- The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other, although they act in concert.

## Definition

composites are materials made by combining two or more natural or artificial elements (with different physical or chemical properties) that are stronger as a team than as individual players. The component materials don't completely blend or lose their individual identities; they combine and contribute their most useful traits to improve the outcome or final product.



# Classification based on reinforcement

***Continuous fiber reinforced composite***

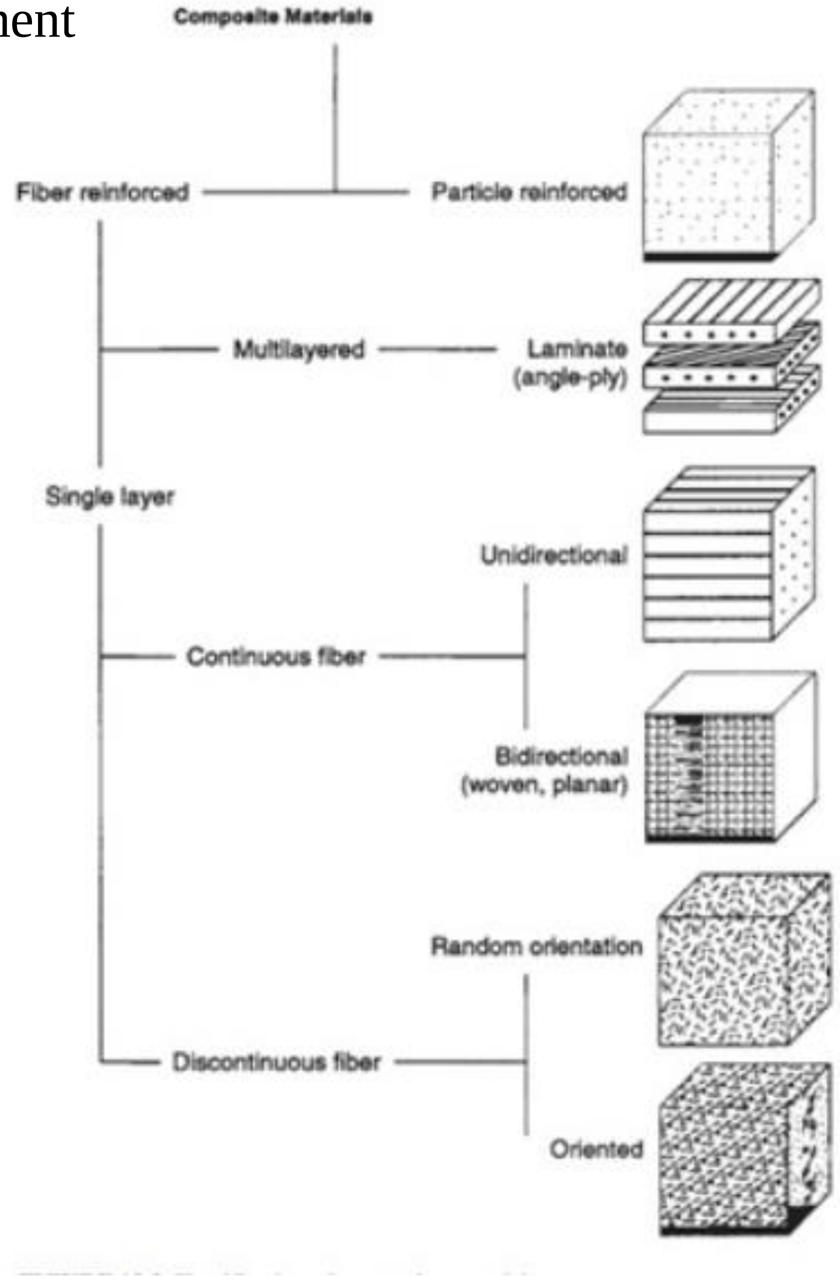
Unidirectional, bidirectional, three directional, multi directional

BI—woven, cross ply, angle ply

***Short fiber/whisker reinforced***

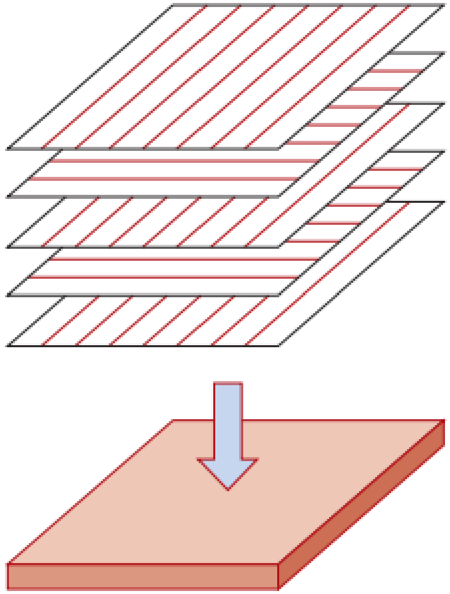
aligned and random

***Particulate composites***

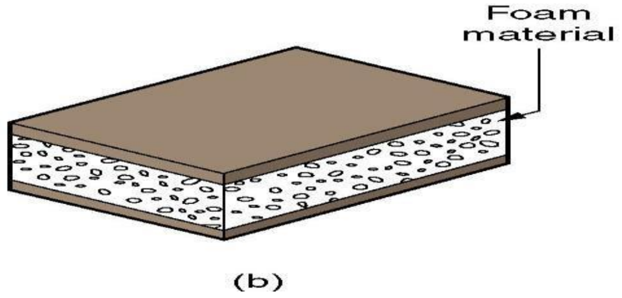


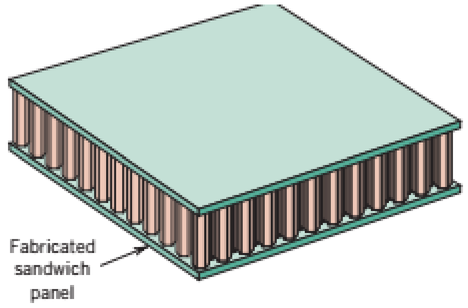
## OTHER COMPOSITE STRUCTURES

**Laminar composite structure :-** *Two or more layers bonded together in an integral piece*



**Sandwich structure:-** *foam & honey cores*



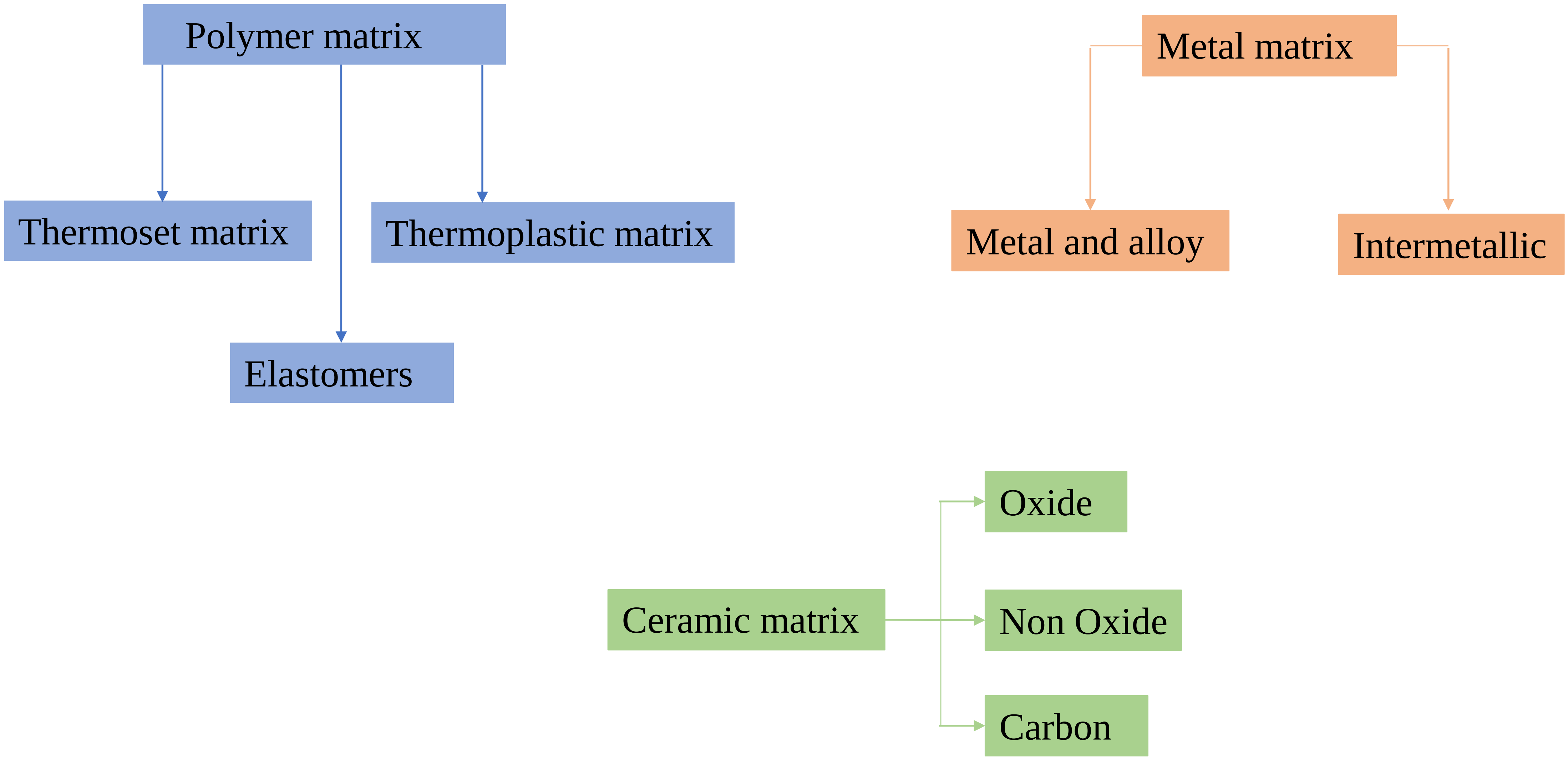



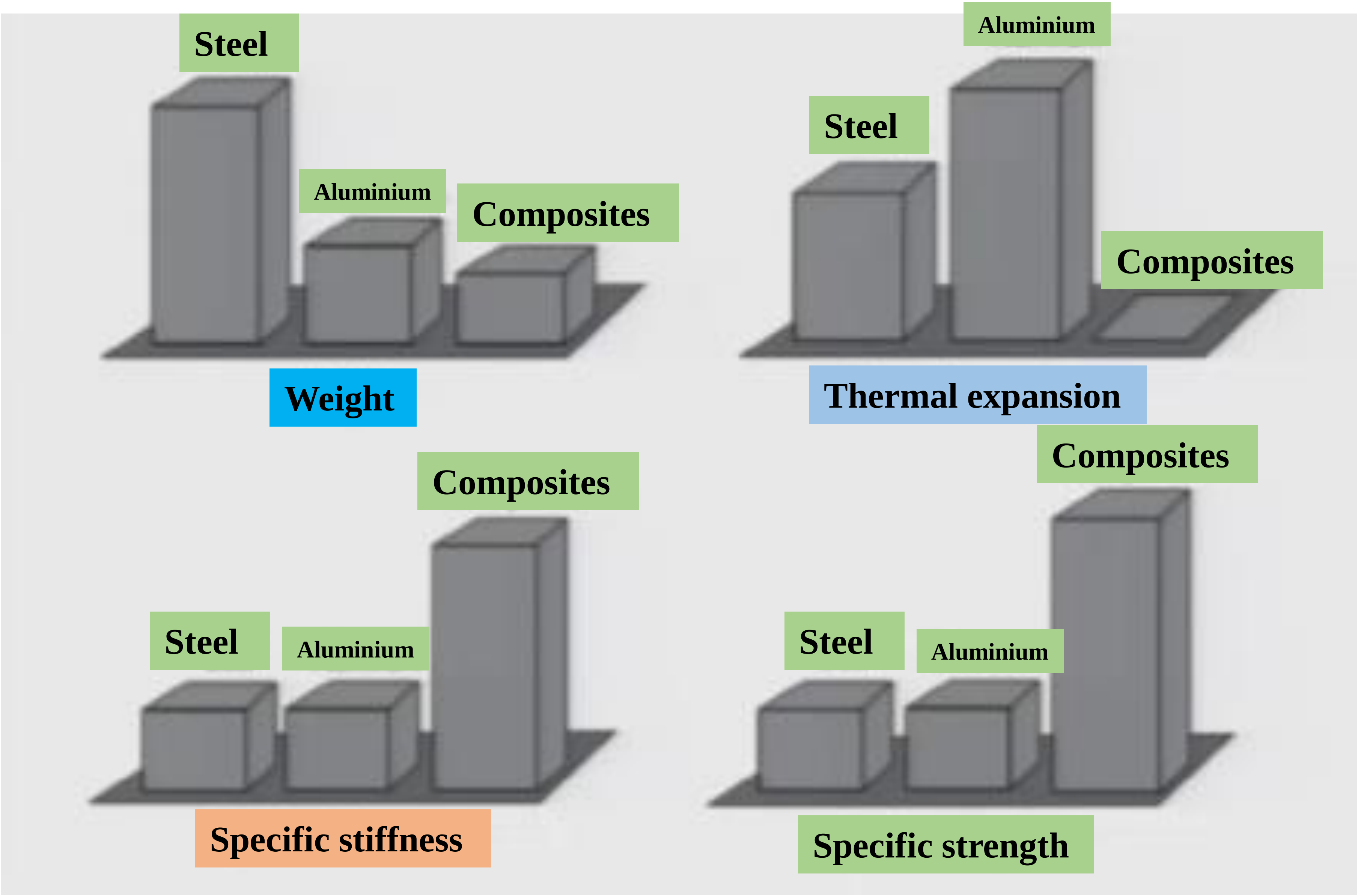

*Consists of a thick core of low density foam bonded on both faces to thin sheets of a different material*

Laminates

- Is composed of two-dimensional sheets or panels that have a preferred high strength direction such as found in wood and continuous and aligned fiber-reinforced plastics.
- The layers are stacked and cemented togethersuch that the orientation of the high-strength
- direction varies with each successive layer.

Classification based on Matrix




Steel
Aluminium
Composites
Weight
Aluminium
Steel
Composites
Thermal expansion
Composites
Steel
Aluminium
Specific stiffness
Composites
Steel
Aluminium
Specific strength

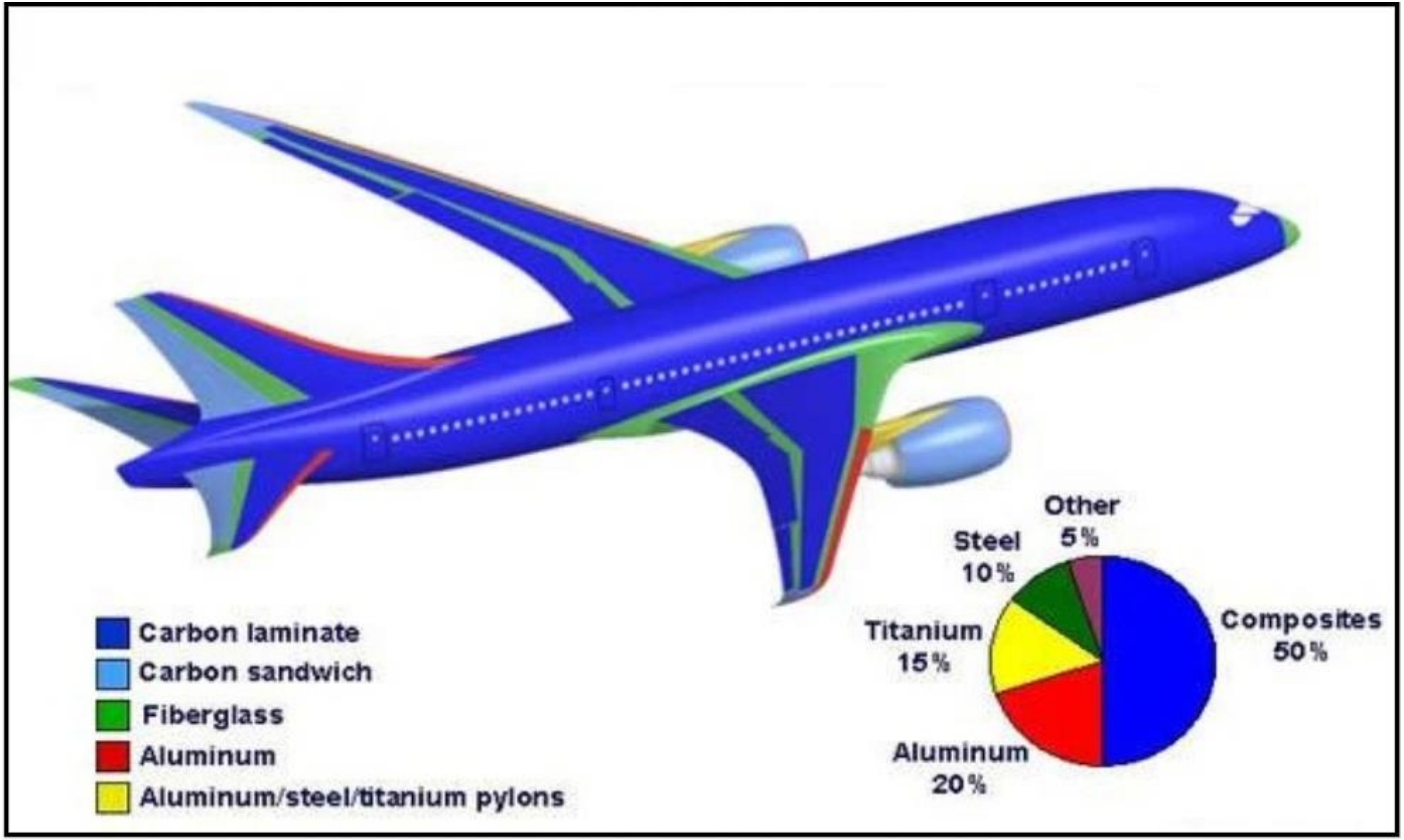
## *Specific features of composites*

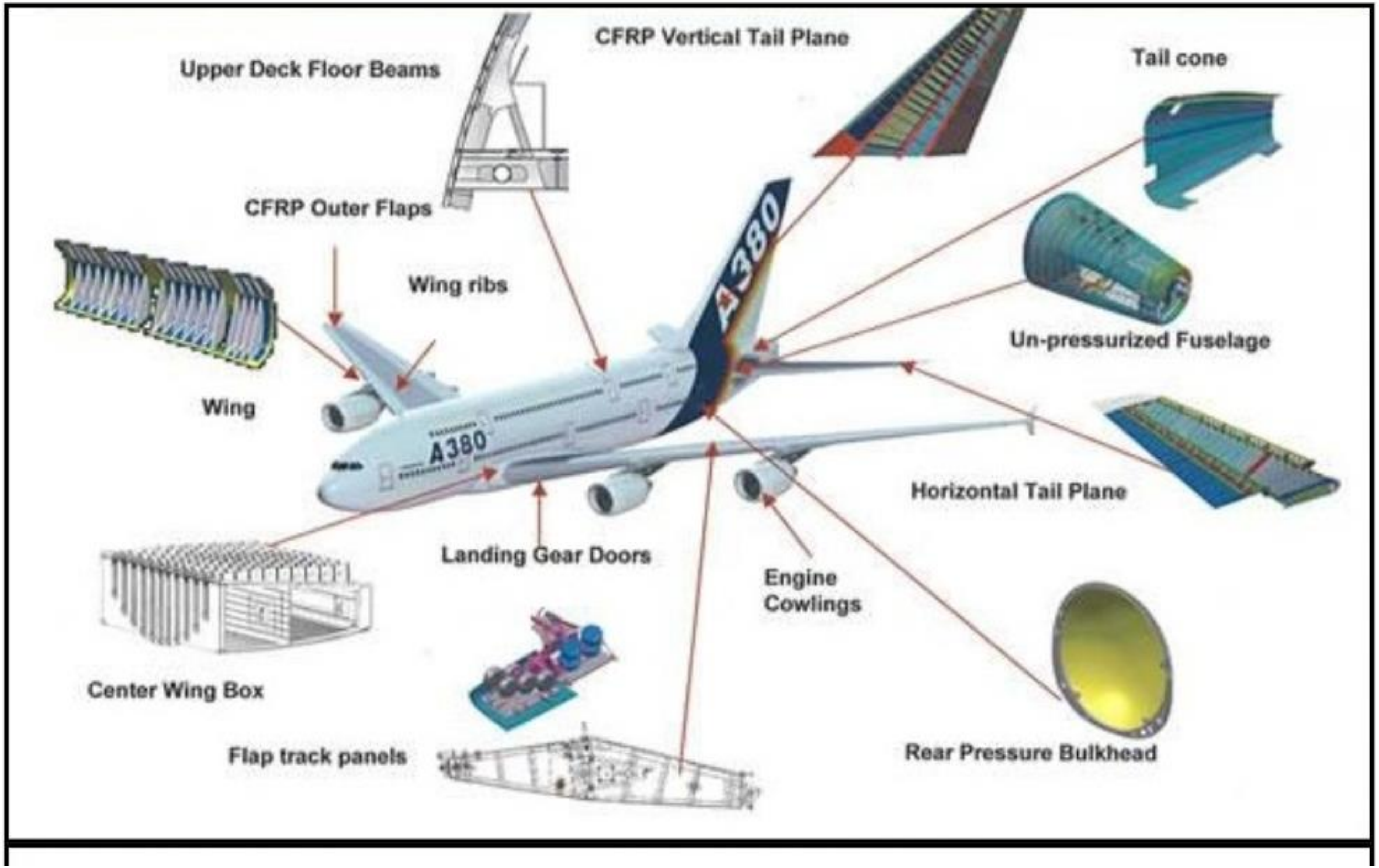
- Composite materials have a high specific stiffness (stiffness-to-density ratio). Composites offer the stiffness of steel at one fifth the weight and equal the stiffness of aluminium at one half the weight.
- The specific strength of a composite material is very high. The specific strength is typically in the range of 3 to 5 times that of steel and aluminium alloy.
- Net shape or near-net shape parts can be produced with composite materials.
- It provides capabilities for part integration. Several metallic components can be replaced by a single composite component.
- Composite structure provide in-service monitoring or online process monitoring with the help of embedded sensors.

## *Composite Applications*

- Engineering/Construction
- Aerospace/Aircraft
- Defense
- Energy
- Sports/Materials
- Ship building/Marine

A Bulletproof Vest


Carbon laminate
Carbon sandwich
Fiberglass
Aluminum
Aluminum/steel/titanium pylons
Other
5%
Steel
10%
Titanium
15%
Composites
50%
Aluminum
20%

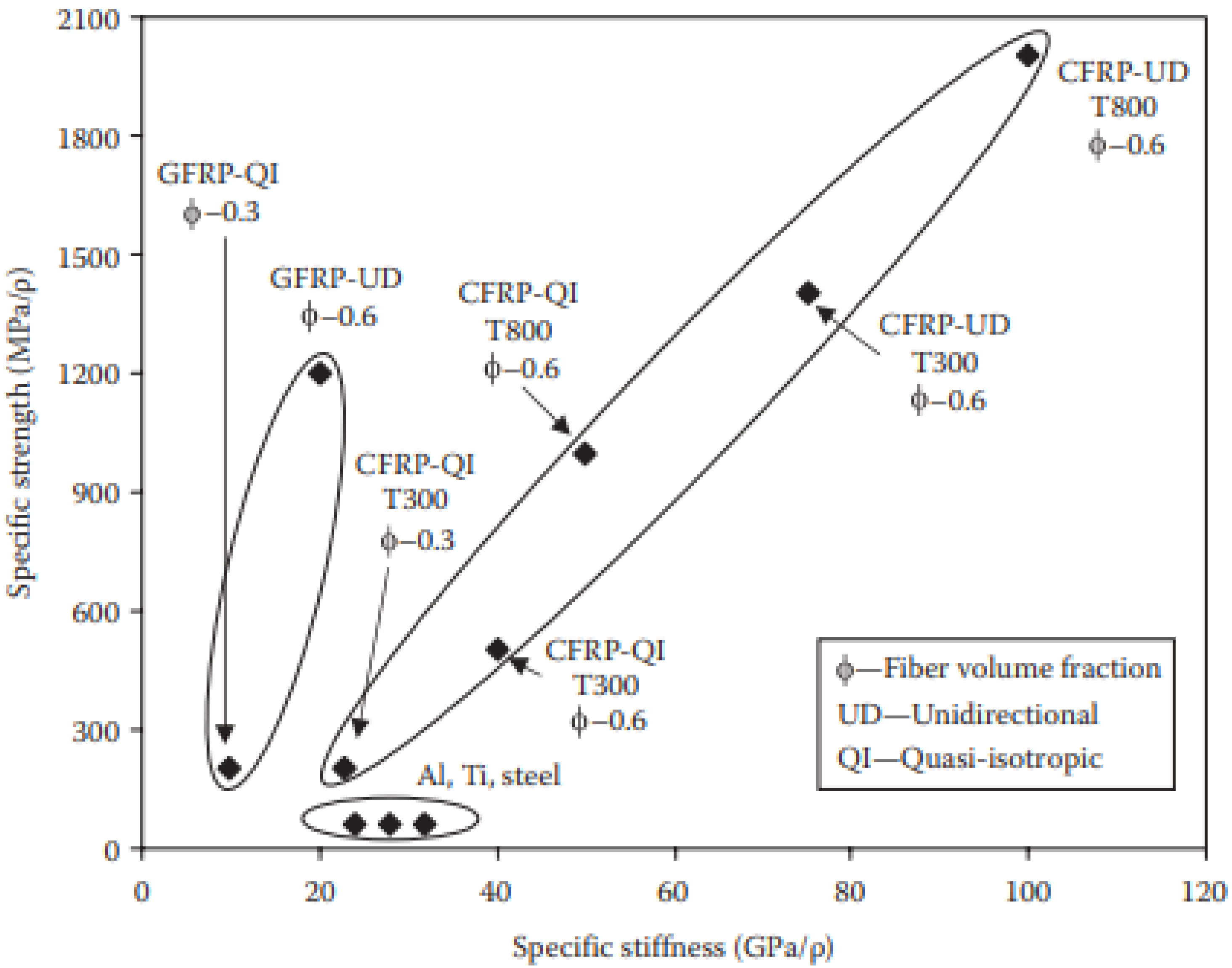


Upper Deck Floor Beams
CFRP Vertical Tail Plane
Tail cone
CFRP Outer Flaps
Wing ribs
Un-pressurized Fuselage
Wing
A380
Horizontal Tail Plane
Landing Gear Doors
Engine Cowlings
Center Wing Box
Flap track panels
Rear Pressure Bulkhead

## ❖ Automotive Industry


Superstructures
Transom
Interior parts
Hulls
Covers
Hatches
Decks
Bulkheads

Specific strength and stiffness values of conventional materials and composites



Matrix

## *Function of matrix*

- The matrix material binds the fibers together and transfer the load to the fibers. It provide s the rigidity and shape to the structure.
- It separates(isolates) fibers from each other, so that individual fibers can act separately
- This stops or slows the propagation of a crack.
- The matrix provides a good surface finish quality and aids in the production of net-shape or near net shape parts.
- Protects the individual fibers from surface damage due to abrasion and oxidation
- Improves fracture toughness of the composite
- Withstand heat or cold, conduct or resist electricity, control chemical attack

## *Types of matrix material and their application*

- Polymer; Metallic; Ceramic

***Polymeric: Commercially, PMCs are more important than MMCs or CMCs***

***Thermoset:***

- Epoxies: Principally used in aerospace applications
- Unsaturated polyesters & vinyl esters: commonly used in automotive, marine, chemical and electrical applications
- Phenolics
- Polyimides: For high temperature aerospace applications (250 – 400 deg C)

***Thermoplastic:***

- ***Aliphatic polyamides(polyesters, polycarbonates, polyacetals) : used with discontinuous fibers in injection-molded parts***
- ***Aromatic polyamide***
- ***PEEK, polysulfone***
- ***PPS, Pei: Suitable for moderately high temperature application with continuous fibers***

| *Thermoset* | *Thermoplastic* |
|---|---|
| ➢ Widely used because composite processing is much easier with low viscosity liquid resin. | ➢ Composite processing is difficult because the viscosity is high even above the melting temperature |
| ➢ Tackiness is a problem | ➢ No tackiness, hence handling is easy |
| ➢ Temperature and pressure requirements are less for the processing of thermoset composites | ➢ Relatively high temperature and pressure are needed for the processing of thermoplastic composites |
| ➢ Unlimited storage life | ➢ Limited storage life |
| ➢ Long curing time, because it involves chemical reaction | ➢ Curing time is lesser |
| ➢ Higher strength and modulus | ➢ Tougher and less brittle |
| ➢ Better thermal stability and chemical resistance | ➢ Lower thermal stability and chemical resistance |
| ➢ Amorphous | ➢ May be semicrystalline |
| ➢ Recycling is difficult | ➢ Can be recycled easily |
| ➢ Post curing often necessary | ➢ Post molding treatment is not recommended, since shrinkage may be severe due to crystallization |

***Thermoset resin***

- Polyester resin
- Epoxy resin
- Polyimide resin
- Phenolic resin

***Polyester resin***

- Low cost resin systems and offer excellent corrosion resistance
- The operating service temperature is lower than for epoxies
- The resin contains a number of c double C  bonds, through which cross links are established.
- An ester is the product of reaction between an acid and alcohol. Similarly unsaturated polyester resin is produced by reacting unsaturated acid/anhydride with a dihydric alcohol

Table 3.2. Some important characteristics of polyester

| Density, $\rho$ (g cm$^{-3}$) | Strength, $\sigma$ (MPa) | Modulus, $E$ (GPa) | Poisson's ratio, $\nu$ | CTE $\alpha$ ($10^{-6}$ K$^{-1}$) | Cure Shrinkage (%) | Use Temp. (°C) |
|---|---|---|---|---|---|---|
| 1.1–1.4 | 30–100 | 2–4 | 0.2–0.33 | 50–100 | 5–12 | 80 |

## *Epoxy resin*

- Epoxy resin is a very versatile resin system, allowing for a broad range of properties and processing capabilities
- It exhibits low shrinkage as well as excellent adhesion to a variety of substrate materials.
- Epoxies are the most widely used resin materials and are used in many applications from aerospace to sporting goods.
- There are varying grades of epoxies with varying levels of performance to meet different application needs.

Table 3.1. Some important characteristics of epoxy

| Density, $\rho$ ($g\ cm^{-3}$) | Strength, $\sigma$ (MPa) | Modulus, $E$ (GPa) | Poisson's ratio, $\nu$ | CTE $\alpha$ ($10^{-6}\ K^{-1}$) | Cure Shrinkage (%) | Use Temp. (°C) |
|---|---|---|---|---|---|---|
| 1.2–1.3 | 50–125 | 2.5–4 | 0.2–0.33 | 50–100 | 1–5 | 150 |

## *ADV:*

- Absence of volatile matters during curing
- Low curing shrinkage
- Excellent resistance to chemicals and solvents
- Excellent adhesion to a wide variety of fillers, fibers and other substances.

## *Polyamide resin*

- These are high temperature polymers, which can be used upto 230 deg C for long periods and upto 315 Deg C for short periods.
- The polymers containing –OC-N-CO- groups are known as polyimides.
- Classsified into Thermosetting and Thermoplastic polyimides.
- Thermoplastic polyimides are derived by condensation reaction between anhydrides and diamines
- The thermosetting resins are derived by addition reaction between unsaturated groups of an amide monomer.

## *ADV:*

- Absence of volatile matters during curing
- Low curing shrinkage
- Excellent resistance to chemicals and solvents
- Excellent adhesion to a wide variety of fillers, fibers and other substances.

## *Thermoplastic resins*

- Nylons
- Propylene (PP)
- Polyethylene terephthalate (PEET)
- Polyetheretherketone (PEEK)
- Polyphenylene sulfide (PPS)

## *Nylons*

- Nylons are used for making intake manifolds, housings, gears, bearings, bushings, sprockets. etc
- Glass filled an carbon filles nylons
- Also called as polyamides. There are several types of nylon, including nylon 6, nylon 66, nylon 11 etc. each offering a variety of mechanical and physical properties.

***Polypropylene***

- These are by far the largest class of synthetic polymers made and used today.
- The attractive properties are low cost of production, light weight and high chemical resistance.
- A wide range of mechanical properties is possible through the use of copolymerization, blending and additives.
- They are stable to oxidation.

***Polyethylene terephthalate***

- It is generally made from dimethyl terephthalate and ethylene glycol.
- It has a high melting point (260 deg C) because of the presence of aromatic ring. It is highly crystalline and rigid.
- The polymer is difficult to process due to the presence of  a large number of aromatic rings in the molecular chain.
- To reduce the crystallinity and rigidity a small amount of dimethyl isophthalate is introduced during the polymerization reaction

***Polyethylene terephthalate***

- PET is one of the most popular commercial polymers. It has very good resistance to many chemicals and has good mechanical strength upto 175 deg C.
- It is widely used to make textiles fibers, films and containers.

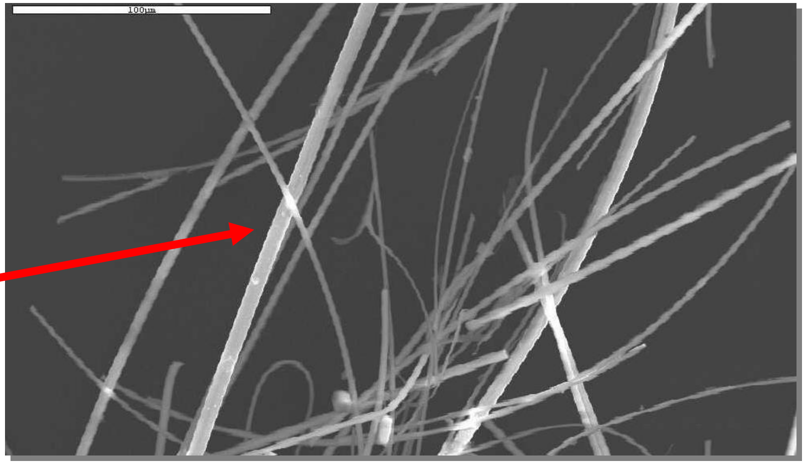
Polyetheretherketone

- It is a new generation thermoplastic that offers the possibility of use at high service temperatures (260 deg C).
- It is a high-performance engineering plastic with outstanding resistance to harsh chemicals, and excellent mechanical strength and dimensional stability.
- PEEK has the advantage of 10 times lower water absorption than epoxies.
- The water absorption of PEEK is 0.5% at room temperature.

***Fibers:***

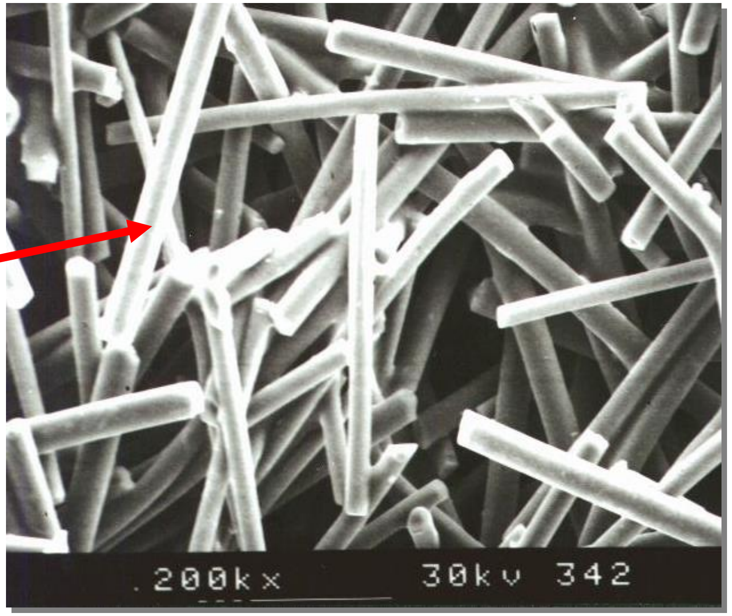
***mainly - glass (GFRPs), carbon (CFRPs), aramid,*** ***others - B, SiC, Al2O3***

***Glass Fiber Reinforced Polymer***



Glass

***Carbon Fiber Reinforced Polymer***




Carbon

# Metal Matrix Composites (MMC`s)

***Metals are very useful matrix materials. They offer following key advantages.***

– High strength

– High modulus

– High toughness

– High impact strength

– Relative immunity to temperature changes

– Immunity to a large range of environmental variables

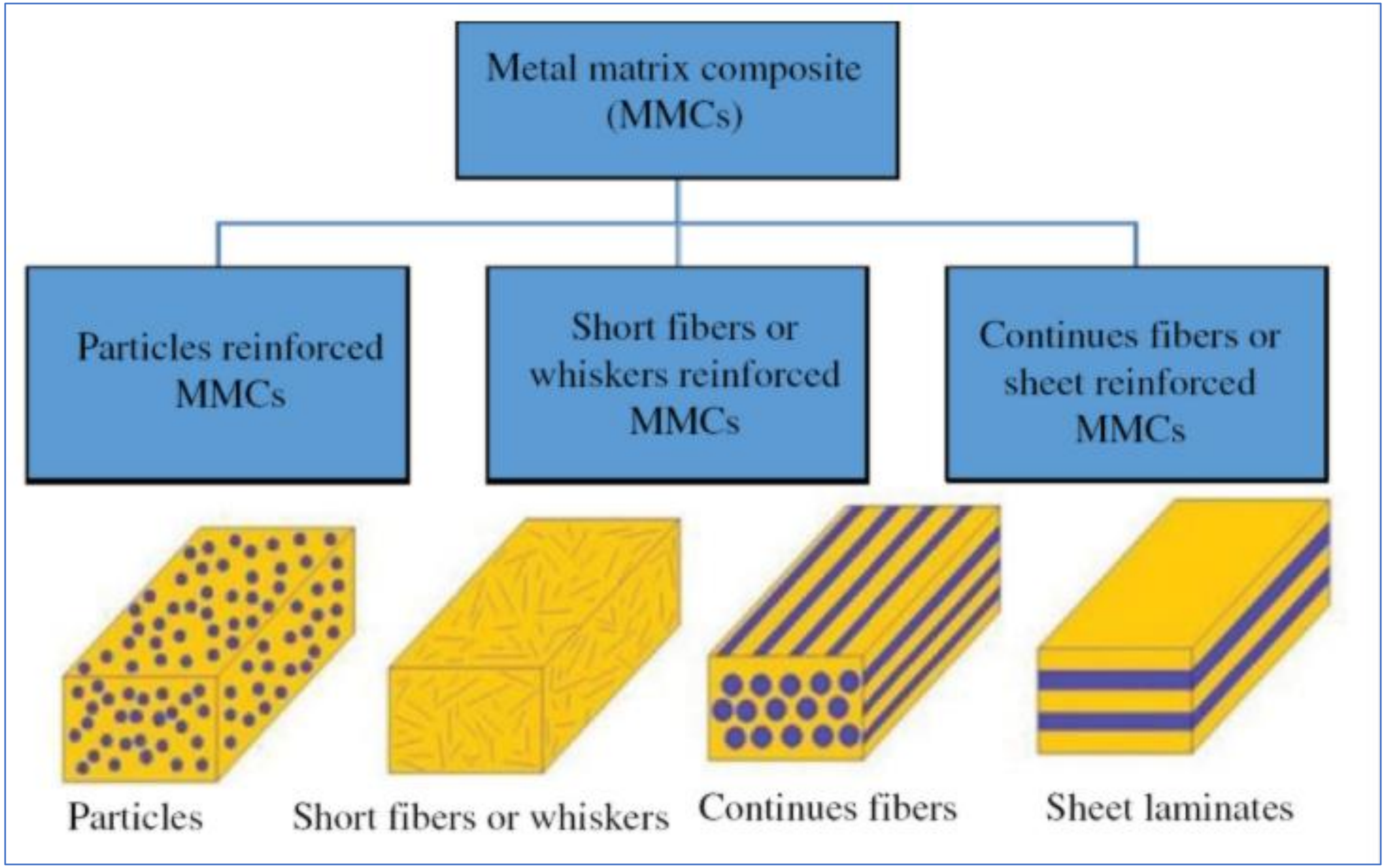
***However, metals are not very popular matrix materials because:***

– They can only be processed at high temperature, as their melting points are high.

– They have high density.

– They have a propensity to react with several types of fibers.

– Metals are susceptible to corrosion



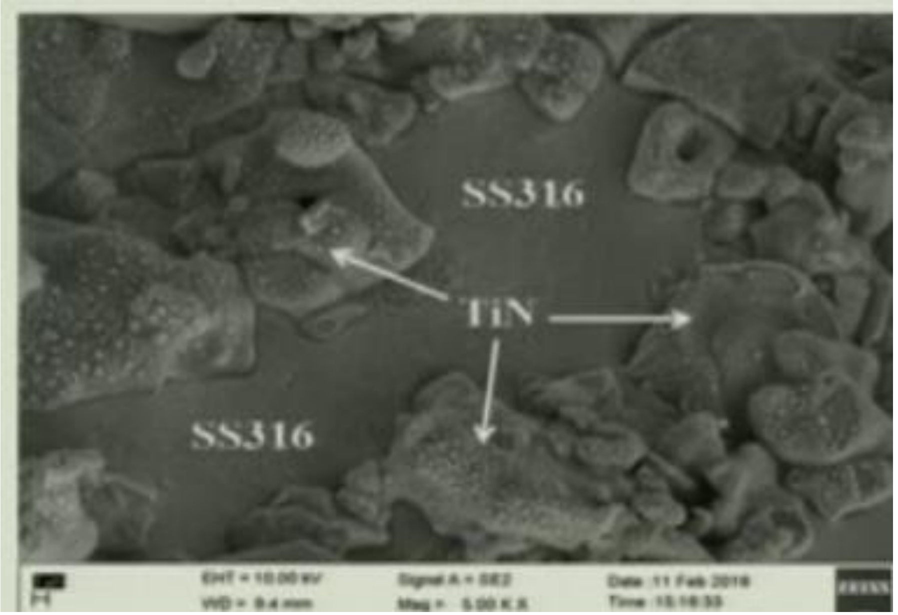

***Matrix:*** Aluminium, Titanium, Magnesium Cobalt, Cobalt Nickel

***Reinforcement:*** Alumina, Zirconia, SiC, TiN

*A MMC is composite material with a least two constituent parts, one being a metal, the other material may be different metal or another material .*

*The matrix material is the monolithic material into which the reinforcement is embedded and it is continuous. In structural applications, the matrix is usually a lighter metal such as Al, Mg or Ti. In high temperature applications, cobalt and cobalt-nickel alloy matrices are common.*

*The reinforcement is embedded into the matrix to enhance or reduce their properties like wear resistance, hardness, density, porosity, mechanical strength, thermal expansion, thermal and electrical conductivities*



## *Reinforcement for MMC*

- ***Low density***
- ***Mechanical compatibility***
- ***Thermal stability***
- ***High young modulus***
- ***High compression and tensile strength***
- ***Good process ability***
- ***Economic efficiency***

**Table 6.1** Typical reinforcements used in metal matrix composites

| Type | Aspect ratio | Diameter (μm) | Examples |
|---|---|---|---|
| Particle | ~1–4 | 1–25 | SiC, $Al_2O_3$, WC, TiC, BN, $B_4C$ |
| Short fiber or whisker | ~10–1,000 | 0.1–25 | SiC, $Al_2O_3$, $Al_2O_3$ + $SiO_2$, C |
| Continuous fiber | >1,000 | 3–150 | SiC, $Al_2O_3$, $Al_2O_3$ + $SiO_2$, C, B, W, NbTi, $Nb_3Sn$ |

## *Different matrices for MMC*

- ***Aluminium Matrix composites***
- ***Magnesium Matric composites***
- ***Titanium Matrix composites***
- ***Cobalt Matric Composites***

### *Aluminium Matrix composites*

- ***Widest group of MMC`s***
- ***Matrices of Aluminium Matrix composites are usually based on Al-Si and on alloys of 2xxx and 6xxx series***
- ***There matrix composites are reinforced by***
- ***Alumina (Al2O3) or Silicon carbide (SiC) in amount of 15-70 Vol%.***
- ***Continuous and discontinuous fibers of Alumina***
- ***Continuous fibers of Silicon Carbide***
- ***Manufactured by the following method***
- ***Powder metallurgy; Stir casting; infiltration***

## *Al-MMC*

*The following properties are the typical for Al-MMC*

- ***High strength at elevated temperature***
- ***High stiffness***
- ***High thermal conductivity***
- ***Excellent abrasion resistance***
- ***It is used for manufacturing automotive parts (pistons, pushrods, brake components) brake rotors for high speed trains, bicycles, golf clubs.***

| *Composite* | *Components* | *Advantages* |
|---|---|---|
| Aluminum–silicon carbide (particle) | Piston | Reduced weight, high strength and wear resistance |
| | Brake rotor, caliper, liner | High wear resistance, reduced weight |
| | Propeller shaft | Reduced weight, high specific stiffness |
| Aluminum–silicon carbide (whiskers) | Connecting rod | Reduced reciprocating mass, high specific strength and stiffness, low coefficient of thermal expansion |
| | Sprockets, pulleys, and covers | Reduced weight, high strength and stiffness |
| Aluminum–aluminum oxide (short fibers) | Piston ring | Wear resistance, high running temperature |
| | Piston crown (combustion bowl) | Reduced reciprocating mass, high creep and fatigue resistance |
| Aluminum–aluminum oxide (long fibers) | Connecting rod | Reduced reciprocating mass, improved strength and stiffness |
| Copper–graphite | Electrical contact strips, electronics packaging, bearings | Low friction and wear, low coefficient of thermal expansion |
| Aluminum–graphite | Cylinder, liner platon, bearings | Call resistance, reduced friction, wear and weight |
| Aluminum–titanium carbide (particle) | Piston, connecting rod | Reduced weight and wear |
| Aluminum–fiber flax | Piston | Reduced weight and wear |
| Aluminum–aluminum oxide fibers–carbon fibers | Engine block | Reduced weight, improved strength and wear resistance |
| Supper alloy-based composite (Ni–$Ni_3Nb$) | Turbine blades | Fatigue resistance, impact strength, temperature resistance |
| Cu–Nb, Cu–$Nb_3Sn$ | Super conductor | Superconducting, mechanical strength, ductility |
| Cu–W | Spot welding electrodes | Burn-up resistance |

# Reinforcement Forms:

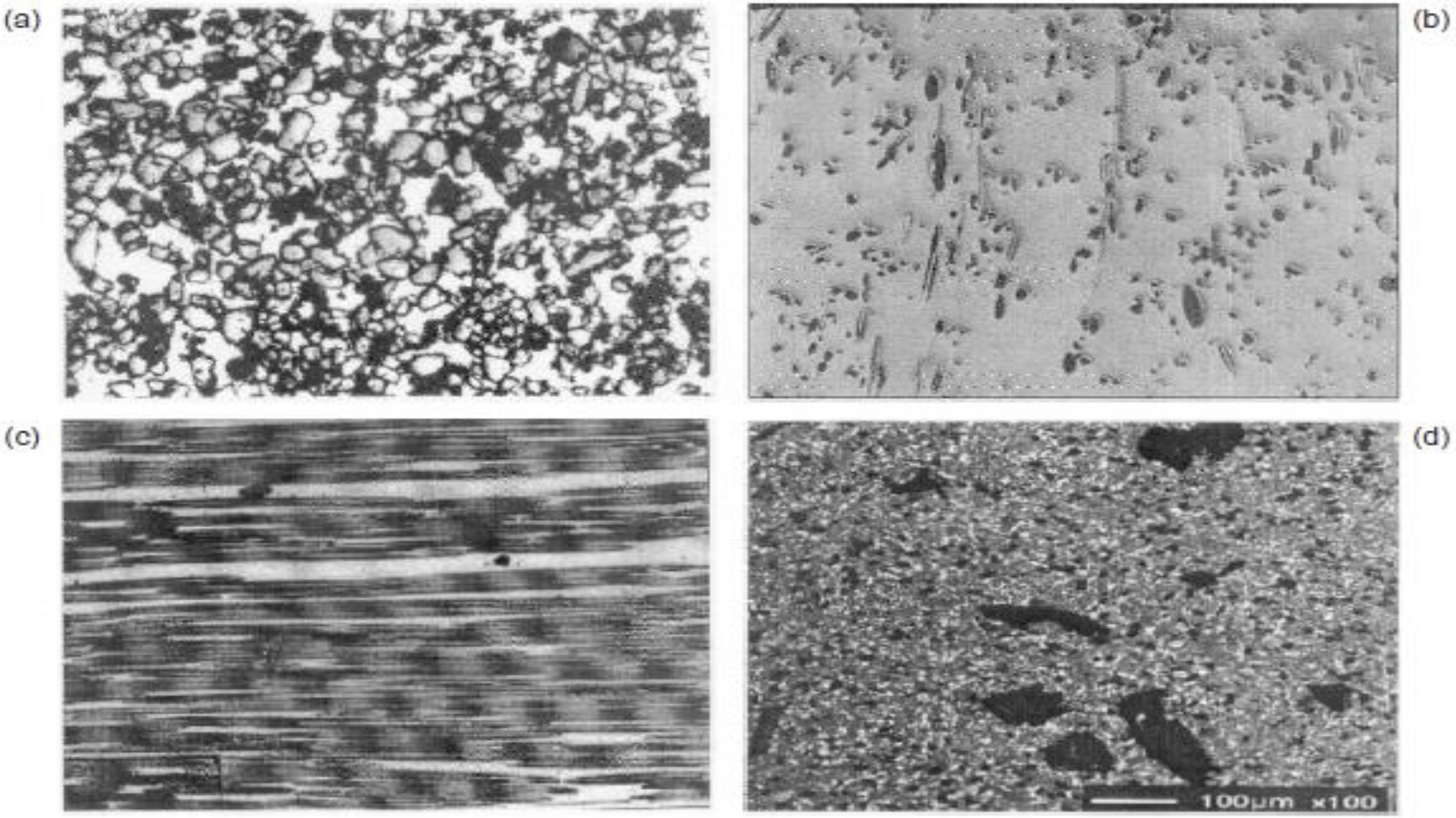



**Figure 1.** Microstructures of **(a)** aluminium matrix composite having high volume fraction of SiC particle reinforcement (40 vol%), **(b)** short fibre-reinforced aluminium matrix composite, **(c)** continuous fibre-reinforced aluminium matrix composite, **(d)** hybrid composite containing 10% SiC and 4% graphite particles.

## *Magnesium Matrix composites*

***They are being reinforced mainly by Sic***

***The following properties are typical for magnesium matrix composite***

- ***Low density***
- ***High stiffness***
- ***High wear resistance***
- ***Good strength even at elevated temperature***
- ***Good creep resistance***

***It is used for manufacturing components for racing cars, light weight, automotive brake system, aircraft parts for gearboxes, transmissions,***

## *Titanium Matrix composites*

***They are being reinforced mainly by***

- ***Continuous SiC***
- ***Titanium boride and titanium Carbide (Particulate composites)***
- ***Titanium matrix composites are manufactured by powder Metallurgy route***

***The following properties are typical for Titanium matrix composite***

- ***High strength***
- ***High stiffness***
- ***High creep resistance***
- ***High thermal stability***
- ***High wear resistance***

***Used for manufacturing structural components of turbine engine components such as fan, blades, rings, shafts, disc), automotive engine components, general machine components.***

.

***Advantage of Metal Matrix composites***

- ***High temperature stability***
- ***Fire resistance***
- ***No moisture absorption***
- ***Higher electrical and thermal conductivities***
- ***Higher specific strength and modulus over metals***
- ***Lower coefficients of thermal expansion than metals by reinforcing with graphite.***

***Disadvantage of Metal Matrix composites***

- ***Higher cost of some material systems.***
- ***Complex fabrication methods.***

## Ceramic matrix composites

- Combination of covalent and ionic bonding between metallic and nonmetallic elements
- High stiffness, low density, chemical inertness, thermal stability, good insulators, etc.
- Operation over a wide range of temperatures
- ***Lack of toughness and brittleness***
- ***catastrophic failure***

***Matrix and reinforcement are in the form of powder***

***Matrix : oxide ceramics, non oxide ceramics***

***Reinforcement:*** continuous, short fibers, ***whiskers, particulates***

***Definition: Ceramic composite are the materials in which one or more ceramic materials are deliberately added to another in order to enhance or provide some property not posses by the original materials.***

***Properties:*** High mechanical strength even at high temperatures; High thermal shock resistance; High stiffness; High toughness; High thermal stability; High corrosion resistance even at high temperatures

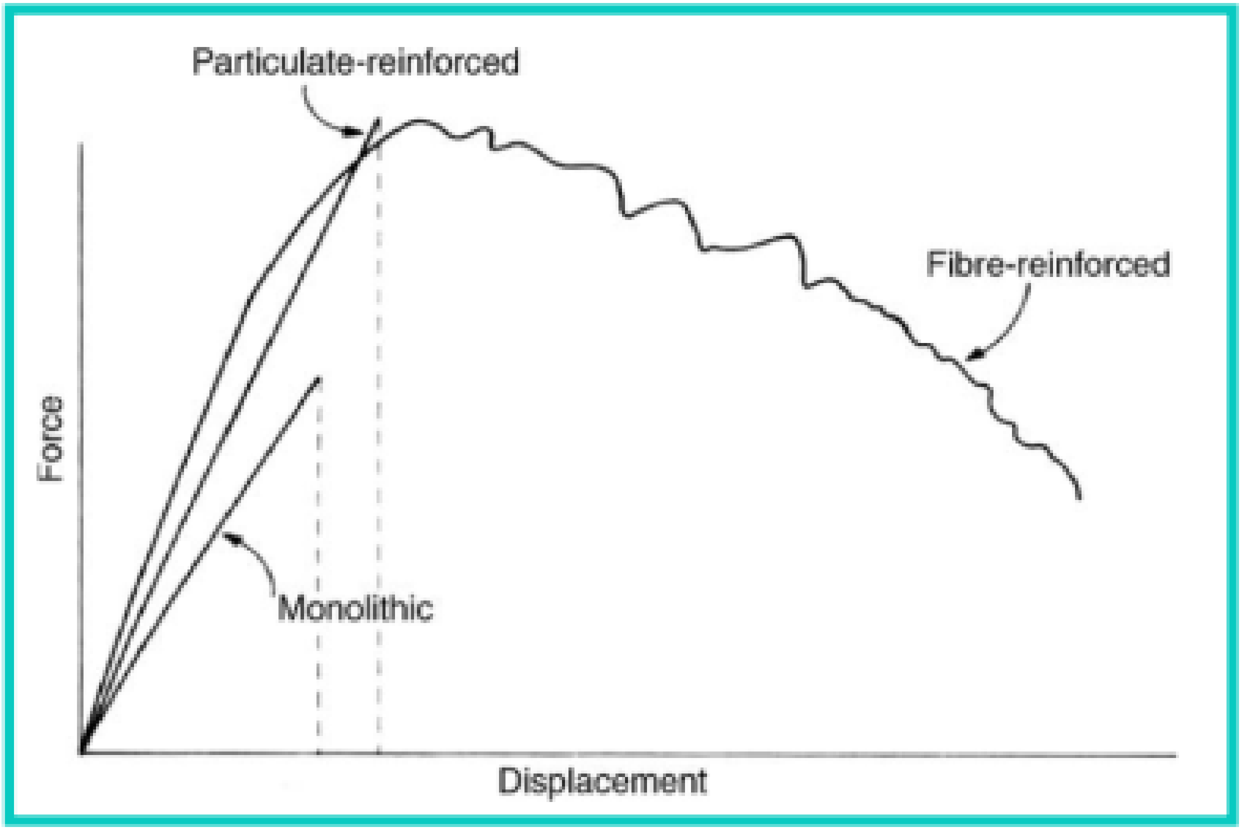
***Most importantly non catastrophic failure.***

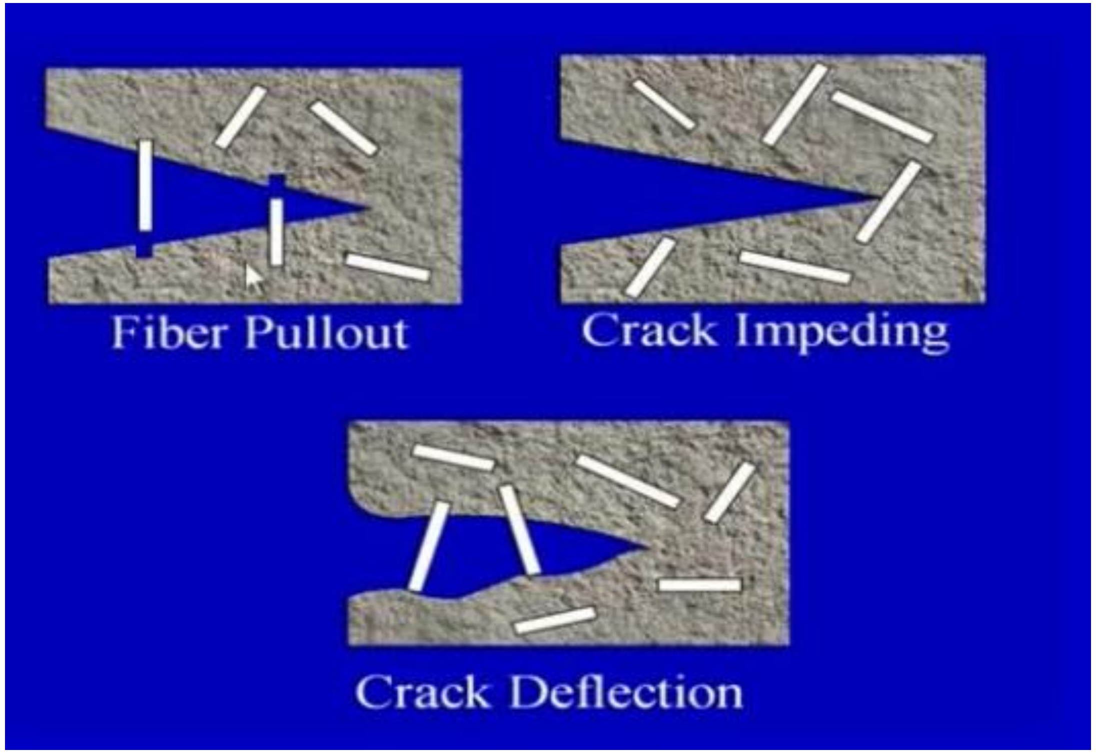
***Processing techniques***: Powder metallurgy. the sintering or hot isostatic pressing.

Infiltration process

Deposition process

Toughening mechanisms



Comparative typical property guide for oxide matrices.

| *Matrix* | *Crystal structure* | *Melting point in pure form,* (°C) | *Onset of reduction in reducing atmospheres,* (°C) | *Theoretical density,* (Mg $m^{-3}$) | *Elastic modulus,* (GPa) | *Typical flexural strength,* (MPa) | *Thermal expansion coefficient,* $10^{-6} K^{-1}$ (25 1000 °C) | *Thermal conductivity,* $W m^{-1} K^{-1}$ (25 °C) | *Electrical conduction*[c] |
|---|---|---|---|---|---|---|---|---|---|
| Alumina, $Al_2O_3$ | Hexagonal | 2060 | 1400 | 3.986 | 400 | 300 | 8 | ≈38 | Insulator |
| Y TZP[a] | Tetragonal | 2300 | 400 (darkens) | 6.10 | 210 | 800 | 10 | ≈1.5 | Insulator/oxygen ion conductor |
| PSZ[b] | Cubic/ tetragonal | 2300 | 400 (darkens) | 6.05 | 200 | 500 | 10 | ≈1.5 | Insulator/oxygen ion conductor |
| Mullite, $3Al_2O_3.2SiO_2$ | Orthorhombic | 1934 | 1000 (darkens) | 3.1 | 300 | 250 | 5 | ≈10 | Insulator |
| Yttrium aluminum garnet (YAG) | Cubic | 2000 | 1400 | | 280 | 250 | 8 | ≈15 | Insulator |
| Spinel, $MgO.Al_2O_3$ | Cubic | 2135 | 1400 | 3.59 | 280 | 250 | 8 | ≈15 | Insulator |
| Na beta alumina | Hexagonal | | | ≈3.5 | 320 | 300 | 8 | | Sodium ion conductor |

[a] TZP tetragonal zirconia polycrystals, containing a high proportion of metastable tetragonal $ZrO_2$, see Section 4.01.5.5. [b] PSZ partially stabilized zirconia, with insufficient stabilizer to form an all-cubic phase material, see Section .5.5. [c] Given in general descriptive terms; actual values are usually strong functions of composition, phase distribution, and temperature. Most oxides are electronic insulators. Those indicated become ionically conducting at raised temperature, typically above 300 400 °C.

Oxide matrices and typical useful reinforcements.

| *Matrix* | *Particulates* | *Platelets* | *Whiskers* | *Fibers* |
|---|---|---|---|---|
| Alumina, $Al_2O_3$ | $ZrO_2$, SiC, TiC, TiN, $TiB_2$, $ZrB_2$, metals | SiC | SiC, $B_4C$ | SiC, $Al_2O_3$, $3Al_2O_3.2SiO_2$ |
| Y TZP[a] | $Al_2O_3$ | $Al_2O_3$ | $Al_2O_3$ | |
| PSZ[b] | $Al_2\ O_3$ | SiC | | |
| Mullite, $3Al_2O_3.2SiO_2$ | $Al_2O_3$, $ZrO_2$ | $Al_2O_3$ | SiC | SiC, $Al_2O_3$ |
| Yttrium aluminum garnet (YAG) | | | $Al_2O_3$ | $Al_2O_3$ |
| Spinel, $MgO.Al_2O_3$ | $Al_2\ O_3$ | | | |
| Na beta alumina | $ZrO_2$ | | | |

[a] TZP tetragonal zirconia polycrystals, containing a high proportion of metastable tetragonal $ZrO_2$,

[b] PSZ partially stabilized zirconia, with insufficient stabilizer to form an all-cubic phase material

Nonoxide matrices other than carbon, and typical reinforcements.

| *Matrix* | *Particulates* | *Platelets* | *Whiskers* | *Fibers* |
|---|---|---|---|---|
| SiC | | SiC | $B_4$ C | C, SiC |
| $Si_3N_4$ | SiC, TiC, TiN, $TiB_2$ | $Si_3N_4$ | $Si_3N_4$ | SiC, $Al_2O_3$, C |
| $MoSi_2$ | $ZrO_2$, $TiB_2$, Nb, SiC, $Si_3N_4$ | $Al_2O_3$ | SiC | SiC, $Al_2O_3$ |
| $B_4C$ | $TiB_2$, $Al_2O_3$ | SiC | SiC | C, SiC |
| BN (hexagonal) | $B_4C$ | | | |
| AlN | | | SiC | |
| $TiB_2$ | $B_4C$, BN | | | |
| TiN[1] | | | | |
| TiC[1] | | | | |

## CARBON–CARBON COMPOSITES

- both reinforcement and matrix are carbon
- Their desirable properties include
- high tensile strengths
- retained to temperatures in excess of 2000 C (3630 F),
- resistance to creep,
- relatively large fracture toughness values.
- low coefficients of thermal expansion and relatively high thermal conductivities;
- these characteristics, coupled with high strengths, give rise to a relatively low susceptibility to thermal shock.
- Their major drawback is a propensity to high temperature oxidation
- employed in rocket motors, as friction materials in aircraft and high-performance automobiles, for hot-pressing molds, in components for advanced turbine engines, and as ablative shields for re-entry vehicles.

1. *What is polymer*
2. *Broad Classification of polymers*
3. *Copolymers, Branched polymers, Side chain polymers*
4. *Polymerization techniques*
5. *Properties and application of polymers*
6. *Amorphous and semicrystalline polymers*
7. *Glass transition temperature and melting temperature in polymers*
8. **Factors determining crystallisation and crystal structure**
9. **Common additives used in polymers.**
10. **Injection molding process**
11. **Thermoforming process**
12. **Compression molding process**
13. **What is composites**
14. **What are the function of matrix and reinforcement**
15. **Classification of composite based on fibers and matrix**
16. **Hand lay up, spray up, Automated layup process for PMC.**

## *Function of Fibers*

1. To carry the load in a structural composite, 70 % to 90% of the load is carried by the fibers.
2. To provide stiffness, strength, thermal stability and other structural properties in the composites
3. To provide electrical conductivity or insulation depending on the type of fiber used.

Factors that governs the fibers contribution are:

- The basic mechanical properties of the fiber itself
- The surface interaction (interface) of the fiber and resin
- The amount of fiber in the composite
- The orientation of the fibers in the composite

## *Types of Fiber*

1. Glass fiber
2. Carbon Fiber
3. Kevlar fiber
4. Boron fiber
5. Natural fiber
6. Hybrid fibers

### *Fiber Spinning Processes*

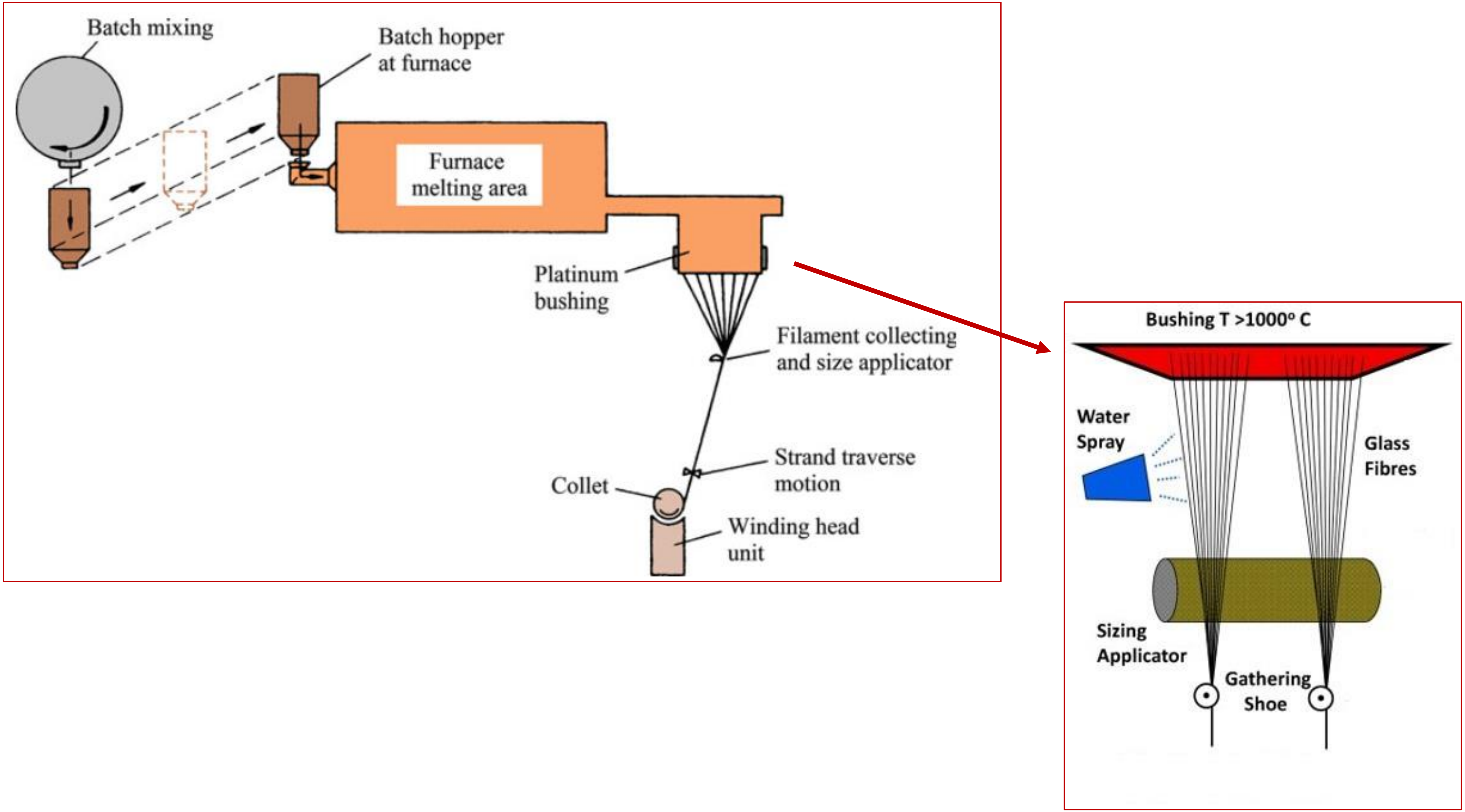
- Wet spinning
- Dry spinning
- Melt spinning

## Glass fibers

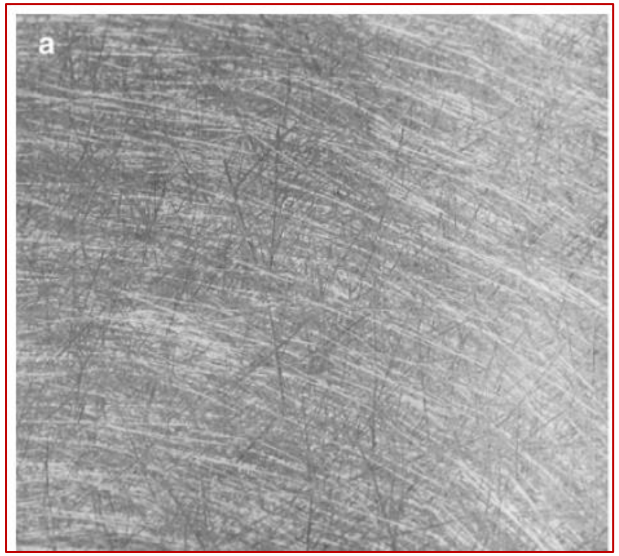
Over 95% of the fibers used in reinforced plastics are glass fibers, as they are inexpensive, easy to manufacture and possess high strength and stiffness with respect to the plastics with which they are reinforced

**Table 2.1** Approximate chemical compositions of some glass fibers (wt.%)

| Composition | E glass | C glass | S glass |
|---|---|---|---|
| $SiO_2$ | 55.2 | 65.0 | 65.0 |
| $Al_2O_3$ | 8.0 | 4.0 | 25.0 |
| $CaO$ | 18.7 | 14.0 | – |
| $MgO$ | 4.6 | 3.0 | 10.0 |
| $Na_2O$ | 0.3 | 8.5 | 0.3 |
| $K_2O$ | 0.2 | – | – |
| $B_2O_3$ | 7.3 | 5.0 | – |


Batch mixing
Batch hopper at furnace
Furnace melting area
Platinum bushing
Filament collecting and size applicator
Strand traverse motion
Collet
Winding head unit
Bushing T >1000° C
Water Spray
Glass Fibres
Sizing Applicator
Gathering Shoe

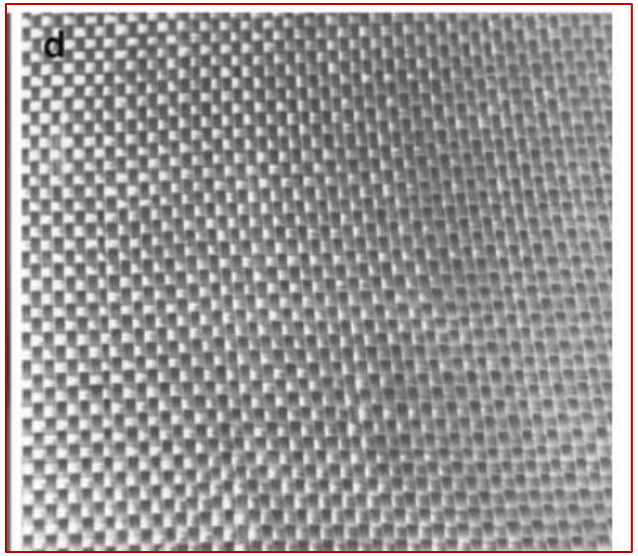
glass fiber is commercially available




(a) chopped strand


(b) continuous yarn


(c) roving




(d) fabric

- Continuous strand is a group of individual fibers;
- Roving is a group of parallel strands;
- Chopped fibers consists of strand or roving chopped to lengths between 5 and 50 mm.
- Glass fibers are also available in the form of oven fabrics or nonwoven mat


Continuous glass fibers

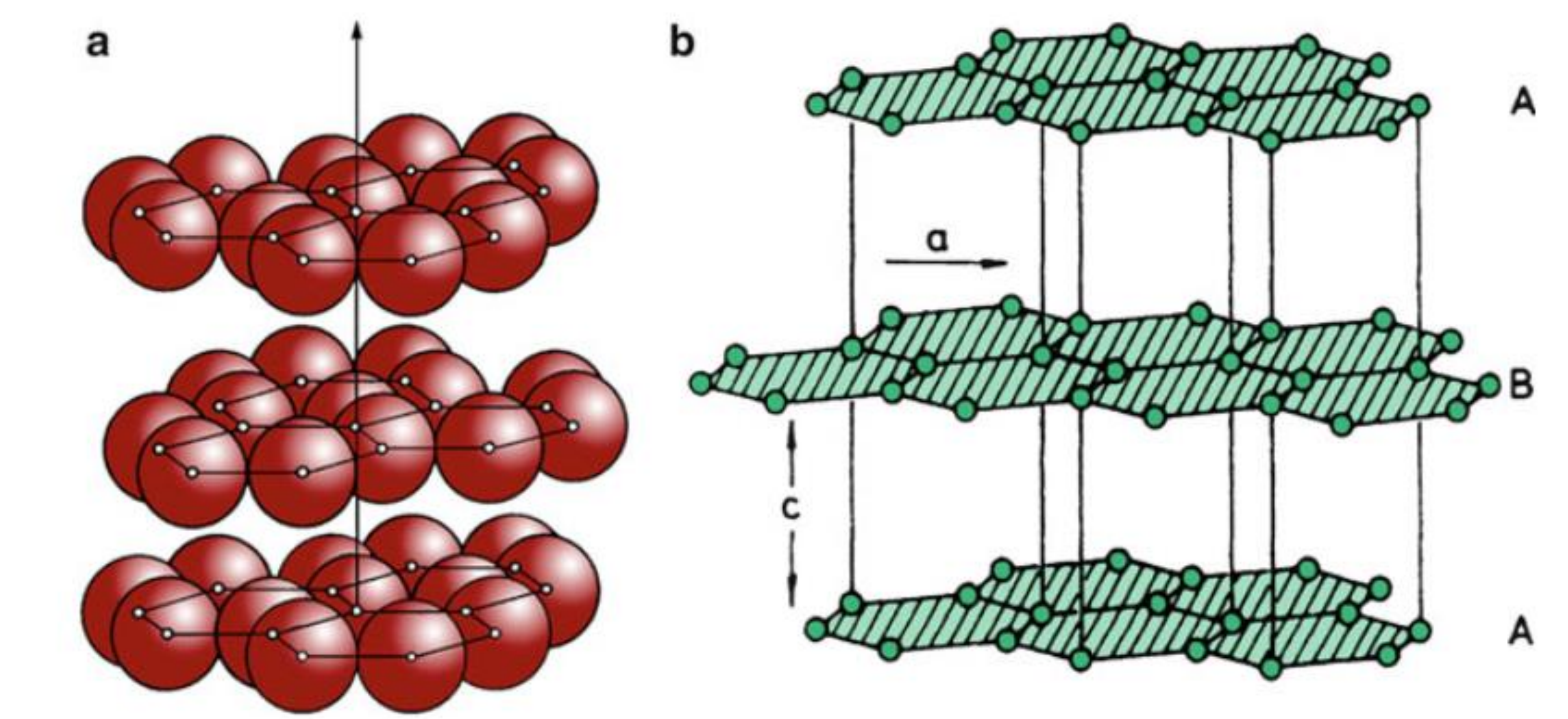
- A sol is a colloidal suspension in which the individual particles are so small (generally in the nm range) that they show no sedimentation.
- A gel, on the other hand, is a suspension in which the liquid medium has become viscous enough to behave more or less like a solid. The sol–gel process of making a fiber involves a conversion of fibrous gels, drawn from a solution at a low temperature, into glass or ceramic fibers at several hundred degrees Celsius.
- The maximum heating temperature in this process is much lower than that in conventional glass fiber manufacture.
- The sol–gel method using metal alkoxides consists of preparing an appropriate homogeneous solution, changing the solution to a sol, gelling the sol, and converting the gel to glass by heating.
- The sol–gel technique is a very powerful technique for making glass and ceramic fibers.

*Fiberglasses (Glass fibers reinforced polymer matrix composites) are characterized by the following properties:*

- High strength-to-weight ratio;
- High modulus of elasticity-to-weight ratio;
- Good corrosion resistance;
- Good insulating properties;
- Low thermal resistance (as compared to metals and ceramics).
- Fiberglass materials are used for manufacturing: boat hulls and marine structures, automobile and truck body panels, pressure vessels, aircraft wings and fuselage sections, housings for radar systems, swimming pools, welding helmets, roofs, pipes.

## *Carbon Fibers*

- density equal to 2.268 g/cm$^3$
- highly anisotropic




(a) Graphitic layer structure.  (b) hexagonal lattice structure of graphite

Most of the carbon fiber fabrication processes involve the following essential steps:

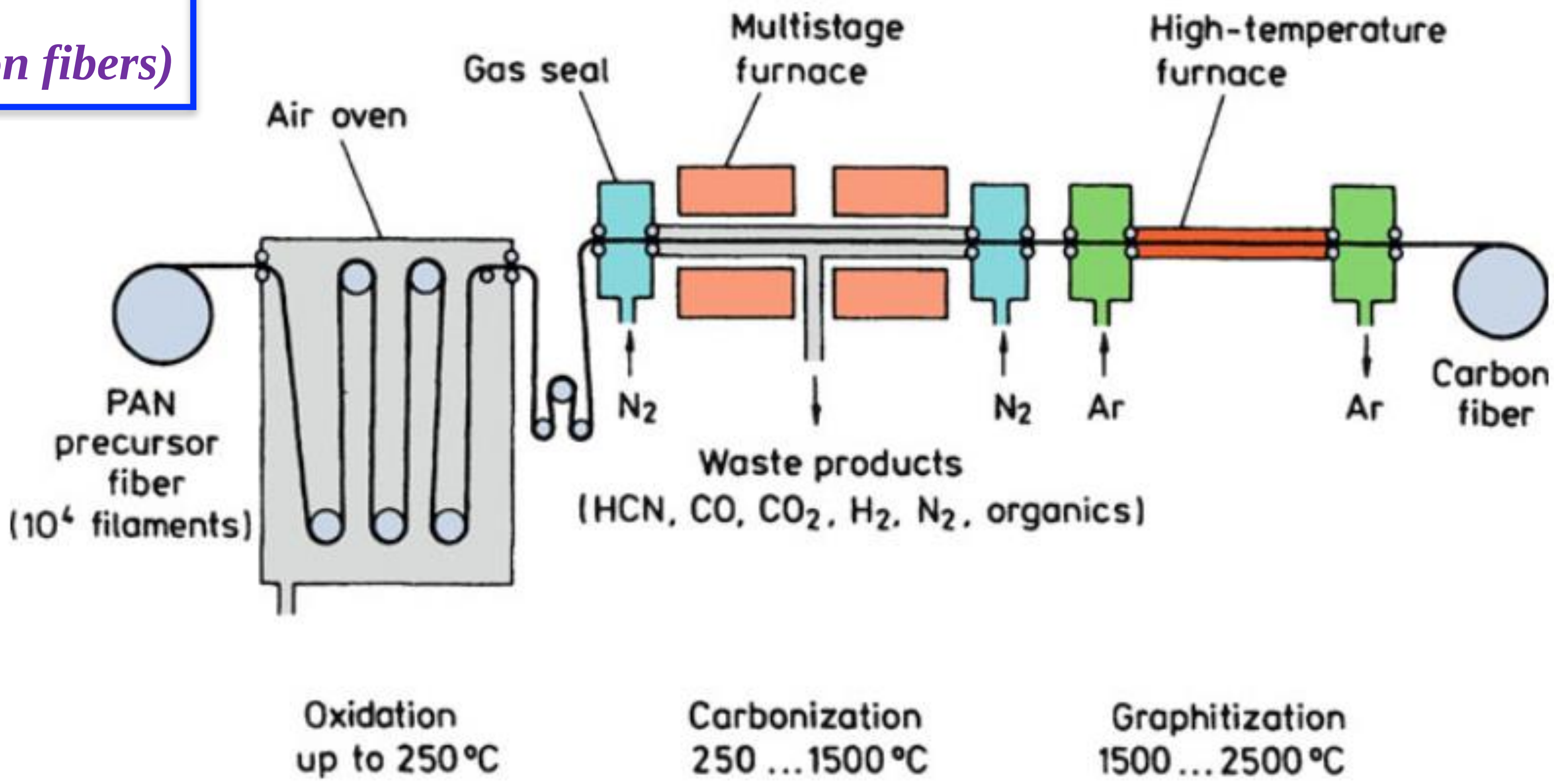
1. A fiberization procedure to make a precursor fiber. This generally involves wet-,dry-, or melt-spinning followed by some drawing or stretching.
2. A stabilization treatment that prevents the fiber from melting in the subsequent high-temperature treatments.
3. A thermal treatment called carbonization that removes most noncarbon elements.
4. An optional thermal treatment called graphitization that improves the properties of carbon fiber obtained in step 3.

*Carbon fibers are also classified according to the manufacturing method:*

- The raw material used to make carbon fiber is called the precursor.
- About 90% of the carbon fibers produced are made from polyacrylonitrile.
- The remaining 10% are made from rayon or petroleum pitch

## *1. PAN-based carbon fibers*

*(the most popular type of carbon fibers)*



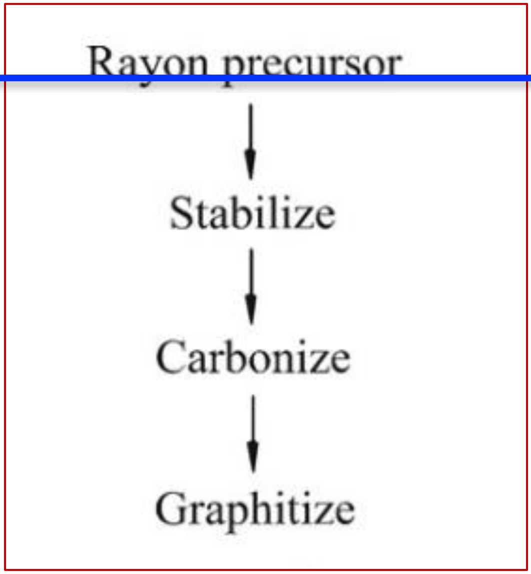
In this method carbon fibers are produced by conversion of polyacrylonitrile (PAN) precursor through the following stages:

- Stretching filaments from polyacrylonitrile precursor and their thermal oxidation at 400°F (200°C). The filaments are held in tension.
- Carbonization in Nitrogen atmosphere at a temperature about 2200 °F (1200°C) for several hours. During this stage non-carbon elements (O,N,H) volatilize resulting in enrichment of the fibers with carbon.
- Graphitization at about 4500 °F (2500°C).

### *2. Rayon-based carbon fiber production*

Rayon is a thermosetting polymer

- fiberization,
- stabilization in a reactive atmosphere (air or oxygen, <400 C),
- carbonization (<1,500 C), and
- graphitization (>2,500 C).




Schematic of rayon-based carbon fiber production

***Carbon Fiber Reinforced Polymers (CFRP) are characterized by the following properties:***

- Light weight; High strength-to-weight ratio;
- Very High modulus elasticity-to-weight ratio;
- High Fatigue strength; Good corrosion resistance;
- Very low coefficient of thermal expansion;
- Low impact resistance; High electric conductivity;
- High cost.
- Carbon Fiber Reinforced Polymers (CFRP) are used for manufacturing: automotive marine and aerospace parts, sport goods (golf clubs, skis, tennis racquets, fishing rods), bicycle frames.

- *The types of carbon fibers are as follows:*
- **UHM** (ultra high modulus). Modulus of elasticity > 65400 ksi (450GPa).
- **HM** (high modulus). Modulus of elasticity is in the range 51000-65400 ksi (350-450GPa).
- **IM** (intermediate modulus). Modulus of elasticity is in the range 29000-51000 ksi (200-350GPa).
- **HT** (high tensile, low modulus). Tensile strength > 436 ksi (3 GPa), modulus of elasticity < 14500 ksi (100 GPa).
- **SHT** (super high tensile). Tensile strength > 650 ksi (4.5GPa).

| Material | Tensile Modulus ($E$) (GPa) | Tensile Strength ($\sigma_u$) (GPa) | Density ($\rho$) ($g/cm^3$) | Specific Modulus ($E/\rho$) | Specific Strength ($\sigma_u/\rho$) |
|---|---|---|---|---|---|
| Fibers | | | | | |
| E-glass | 72.4 | 3.5[a] | 2.54 | 28.5 | 1.38 |
| S-glass | 85.5 | 4.6[a] | 2.48 | 34.5 | 1.85 |
| Graphite (high modulus) | 390.0 | 2.1 | 1.90 | 205.0 | 1.1 |
| Graphite (high tensile strength) | 240.0 | 2.5 | 1.90 | 126.0 | 1.3 |
| Boron | 385.0 | 2.8 | 2.63 | 146.0 | 1.1 |
| Silica | 72.4 | 5.8 | 2.19 | 33.0 | 2.65 |
| Tungsten | 414.0 | 4.2 | 19.30 | 21.0 | 0.22 |
| Beryllium | 240.0 | 1.3 | 1.83 | 131.0 | 0.71 |
| Kevlar 49 (aramid polymer) | 130.0 | 2.8 | 1.50 | 87.0 | 1.87 |
| Conventional materials | | | | | |
| Steel | 210.0 | 0.34–2.1 | 7.8 | 26.9 | 0.043–0.2 |
| Aluminum alloys | 70.0 | 0.14–0.62 | 2.7 | 25.9 | 0.052–0.2 |

***Kevlar - aramid fibers.***

- Kevlar fibers were originally developed as a replacement of steel in automotive tires.
- Kevlar may protect carbon fibers and improve their properties: hybrid fabric (Kevlar + Carbon fibers) combines very high tensile strength with high impact and abrasion resistance.
- High tensile strength (five times stronger per weight unite than steel);
- High modulus of elasticity; Very low elongation up to breaking point; Low weight; High chemical inertness;
- Very low coefficient of thermal expansion; High Fracture Toughness (impact resistance);
- High cut resistance; Textile processibility; Flame resistance.
- The disadvantages of Kevlar are: ability to absorb moisture (making Kevlar composites more sensitive to the environment), difficulties in cutting, low compressive strength.

**Table 2.6** Properties of Kevlar aramid fiber yarns[a]

| Property | K 29 | K 49 | K 119 | K 129 | K 149 |
|---|---|---|---|---|---|
| Density ($g/cm^3$) | 1.44 | 1.45 | 1.44 | 1.45 | 1.47 |
| Diameter (μm) | 12 | 12 | 12 | 12 | 12 |
| Tensile strength (GPa) | 2.8 | 2.8 | 3.0 | 3.4 | 2.4 |
| Tensile strain to fracture (%) | 3.5–4.0 | 2.8 | 4.4 | 3.3 | 1.5–1.9 |
| Tensile modulus (GPa) | 65 | 125 | 55 | 100 | 147 |
| Moisture regain (%) at 25 °C, 65 % RH | 6 | 4.3 | – | – | 1.5 |
| Coefficient of expansion ($10^{-6}$ $K^{-1}$) | −4.0 | −4.9 | – | – | – |

[a]All data from Du Pont brochures. Indicative values only. 25-cm yarn length was used in tensile tests (ASTM D-885). *K* stands for Kevlar, a trademark of Du Pont

Kevlar. This is meant mainly for use as rubber reinforcement for tires (belts or radial tires for cars and carcasses of radial tires for trucks)

Kevlar 29. This is used for ropes, cables, coated fabrics for inflatables, architectural fabrics, and ballistic protection fabrics. Vests made of Kevlar 29 have been used by law-enforcement agencies in many countries.

Kevlar 49. This is meant for reinforcement of epoxy, polyester, and other resins for use in the aerospace, marine, automotive, and sports industries.

# TYPICAL PROPERTIES OF STRUCTURAL FIBERS

| Fiber Type | Density ($kg/m^3$) | E-Modulus (GPa) | Tensile Strength (GPa) | Elong. (%) |
|---|---|---|---|---|
| E-Glass | 2.54 | 72.5 | 1.72-3.45 | 2.5 |
| S-Glass | 2.49 | 87 | 2.53-4.48 | 2.9 |
| Kevlar 29 | 1.45 | 85 | 2.27-3.80 | 2.8 |
| Kevlar 49 | 1.45 | 117 | 2.27-3.80 | 1.8 |
| Carbon (HS) | 1.80 | 227 | 2.80-5.10 | 1.1 |
| Carbon (HM) | 1.80-1.86 | 370 | 1.80 | 0.5 |
| Carbon (UHM) | 1.86-2.10 | 350-520 | 1.00-1.75 | 0.2 |

# ADVANTAGES AND DISADVANTAGES OF REINFORCING FIBERS

| Fiber Type | Advantages | Disadvantages |
|---|---|---|
| E-Glass, S-Glass | High Strength, Low Cost | Low Stiffness, Fatigue |
| Aramid | High Strength, Low Density | Low Compr. Str., High Moisture Absorption |
| HS Carbon | High Strength and Stiffness | High Cost |
| UHM Carbon | Very High Stiffness | Low Strength, High Cost |